Client-use note. Prepared as prospect-facing GTM collateral from representative engineering datasets and cited references. Class-typical vessel values are not a substitute for owner-approved vessel data, project-specific metocean, certified RAOs, or final installation engineering.

DIGITALMODEL — ENGINEERING INTELLIGENCE

Deepwater Mudmat Installation Analysis

180 parametric cases: 2 vessels x 6 depths x 3 mudmats x 5 sea states

180 parametric cases analysed overnight

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Methodology

This analysis screens deepwater mudmat installations across a parametric matrix of vessels, water depths, structure sizes, and sea states. Each case evaluates five installation phases per DNV-RP-H103 and DNV-ST-N001 methodology.

Phase 1: Lift-off

Hook load = $mass_{air} \times g \times DAF_{lift}$ (1.10). Checked against crane SWL at operating radius via interpolation of the crane capacity curve.

Phase 2: In-air Transit

Tilt check for 4-point lift with spreader beam. CoG offset / hook height gives tilt angle, checked against 5° limit. Symmetric mudmats always pass.

Phase 3: Splash Zone (Critical)

Slamming force: $F_{slam} = 0.5 \times \rho \times C_s \times A_p \times v_{rel}^2$, where $v_{rel} = v_{lowering} + v_{wave}$. Additional varying buoyancy and drag forces. DAF_{splash} = 1.30. This phase typically governs for large mudmats in higher sea states.

Phase 4: Lowering through Water Column

Cable tension = $W_{sub} + W_{cable}(depth)$ + snap load. Snap load from dynamic cable response using vessel heave RAO. Checked against 85% of wire MBL. Becomes critical at extreme depths due to cable self-weight.

Phase 5: Landing

Bearing pressure = W_{sub} / A_{base}. Checked against 50 kPa soft clay bearing capacity. Rarely governs for typical mudmat proportions.

Go/No-Go Criteria

- **GO:** All phases pass AND max utilisation < 0.85
- **MARGINAL:** All phases pass AND max utilisation in [0.85, 1.00]
- **NO-GO:** Any phase fails (utilisation > 1.00)

Applicable Codes & Standards

- DNV-RP-H103 (2011) — Modelling and Analysis of Marine Operations
- DNV-ST-N001 (2021) — Marine Operations and Marine Warranty
- DNV-RP-C205 (2010) — Environmental Conditions and Environmental Loads

Installation Screening Summary

All cases at Hs = 2.0 m reference sea state

VESSEL	STRUCTURE	DEPTH (M)	MAX UTIL.	GOVERNING PHASE	STATUS
Large CSV	Mudmat-S-50te	500	24.7%	Landing	GO
Large CSV	Mudmat-M-100te	500	21.5%	Landing	GO

VESSEL	STRUCTURE	DEPTH (M)	MAX UTIL.	GOVERNING PHASE	STATUS
Large CSV	Mudmat-L-200te	500	20.6%	Splash zone	GO
Large CSV	Mudmat-S-50te	1000	24.7%	Landing	GO
Large CSV	Mudmat-M-100te	1000	21.5%	Landing	GO
Large CSV	Mudmat-L-200te	1000	20.6%	Splash zone	GO
Large CSV	Mudmat-S-50te	1500	24.7%	Landing	GO
Large CSV	Mudmat-M-100te	1500	21.5%	Landing	GO
Large CSV	Mudmat-L-200te	1500	20.6%	Splash zone	GO
Large CSV	Mudmat-S-50te	2000	24.7%	Landing	GO
Large CSV	Mudmat-M-100te	2000	21.5%	Landing	GO
Large CSV	Mudmat-L-200te	2000	20.6%	Splash zone	GO
Large CSV	Mudmat-S-50te	2500	24.7%	Landing	GO
Large CSV	Mudmat-M-100te	2500	21.5%	Landing	GO
Large CSV	Mudmat-L-200te	2500	20.6%	Splash zone	GO
Large CSV	Mudmat-S-50te	3000	24.7%	Landing	GO
Large CSV	Mudmat-M-100te	3000	21.5%	Landing	GO
Large CSV	Mudmat-L-200te	3000	20.6%	Splash zone	GO
Medium CSV	Mudmat-S-50te	500	24.7%	Landing	GO
Medium CSV	Mudmat-M-100te	500	34.8%	Splash zone	GO
Medium CSV	Mudmat-L-200te	500	71.5%	Splash zone	GO
Medium CSV	Mudmat-S-50te	1000	24.7%	Landing	GO
Medium CSV	Mudmat-M-100te	1000	34.8%	Splash zone	GO
Medium CSV	Mudmat-L-200te	1000	71.5%	Splash zone	GO
Medium CSV	Mudmat-S-50te	1500	24.7%	Landing	GO
Medium CSV	Mudmat-M-100te	1500	34.8%	Splash zone	GO
Medium CSV	Mudmat-L-200te	1500	71.5%	Splash zone	GO
Medium CSV	Mudmat-S-50te	2000	24.7%	Landing	GO
Medium CSV	Mudmat-M-100te	2000	34.8%	Splash zone	GO
Medium CSV	Mudmat-L-200te	2000	71.5%	Splash zone	GO
Medium CSV	Mudmat-S-50te	2500	24.7%	Landing	GO
Medium CSV	Mudmat-M-100te	2500	34.8%	Splash zone	GO
Medium CSV	Mudmat-L-200te	2500	71.5%	Splash zone	GO
Medium CSV	Mudmat-S-50te	3000	24.7%	Landing	GO
Medium CSV	Mudmat-M-100te	3000	34.8%	Splash zone	GO
Medium CSV	Mudmat-L-200te	3000	71.5%	Splash zone	GO

LIVE

Take This Analysis Live During Operations

This report used **design sea states** to screen parametric cases overnight. During the actual operation, **digitalmodel** can feed your vessel's measured motion data — VMMS, IMMS, MRU — directly into the same engineering models for **real-time go/no-go decisions**.

Instead of relying on forecasted Hs limits, the installation screening updates continuously with actual crane tip motions, hook loads, and dynamic amplification factors measured on your vessel.

$500K VC-funded — we are building overnight engineering and real-time operations support exclusively for marine installation contractors. Early adopters get priority onboarding and custom model calibration for their fleet.

Schedule a Technical Demo →

Assumptions & Limitations

- Crane SWL evaluated at maximum capacity radius (most favourable position)
- DAF lift-off = 1.10, DAF splash zone = 1.30 per DNV-ST-N001
- JONSWAP peak period approximation: Tp = 4 * sqrt(Hs)
- Slamming coefficient Cs = 5.0 for flat-bottom structures per DNV-RP-H103 Table 4-1
- Added mass coefficient Ca = 1.0, drag coefficient Cd = 2.0 for flat plate geometry
- Wire rope MBL safety factor = 0.85 (max tension / MBL)
- Bearing capacity limit = 50 kPa (soft clay, undrained)
- Vessel heave at crane tip uses simplified single-frequency RAO peak
- No current loads or wind loads on structure during lowering
- Rigging weight excluded from hook load calculations
- Seabed assumed flat — no slope corrections for landing
- Seawater density = 1025 kg/m3 throughout water column

Chart 1: Go/No-Go Installation Matrix

Side-by-side vessel comparison at Hs = 2.0 m reference sea state.

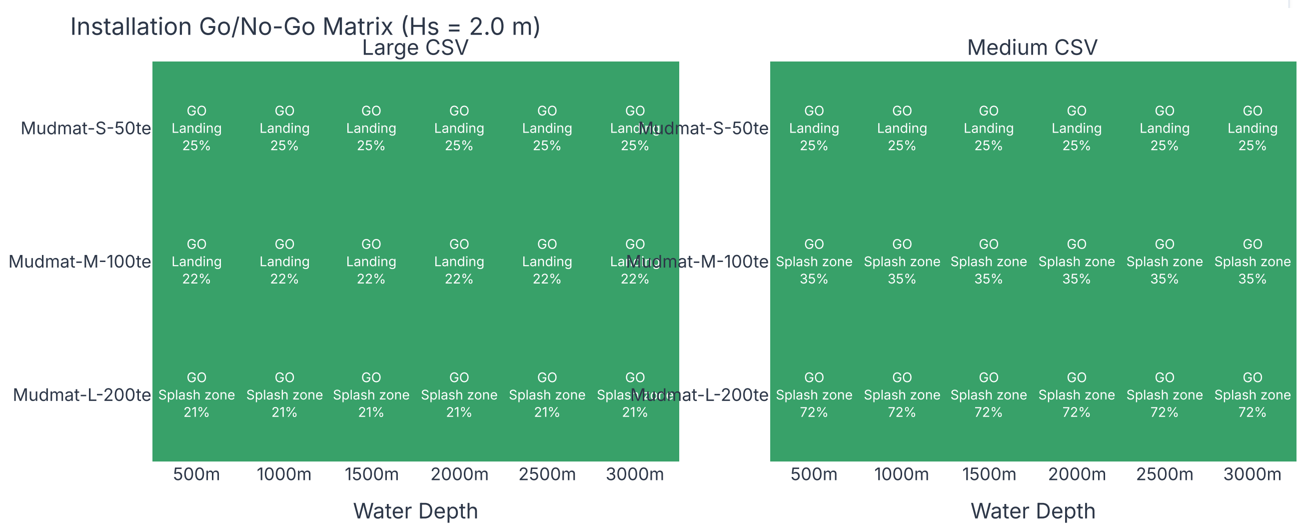


Chart 2: Crane Utilisation vs Water Depth

Max utilisation across all phases. Solid = Large CSV, dashed = Medium CSV.

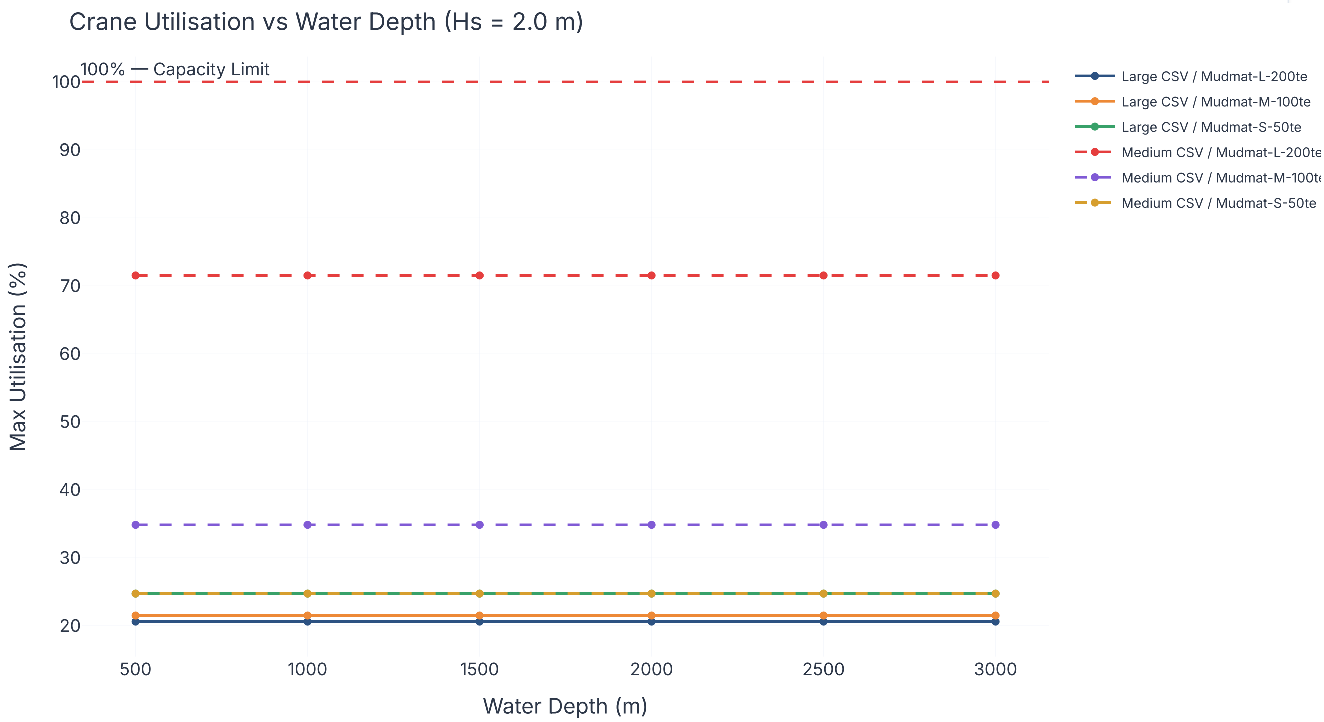


Chart 3: Phase Utilisation vs Water Depth — Splash & Lowering

Shows how splash zone dominates shallow cases while lowering governs at depth.

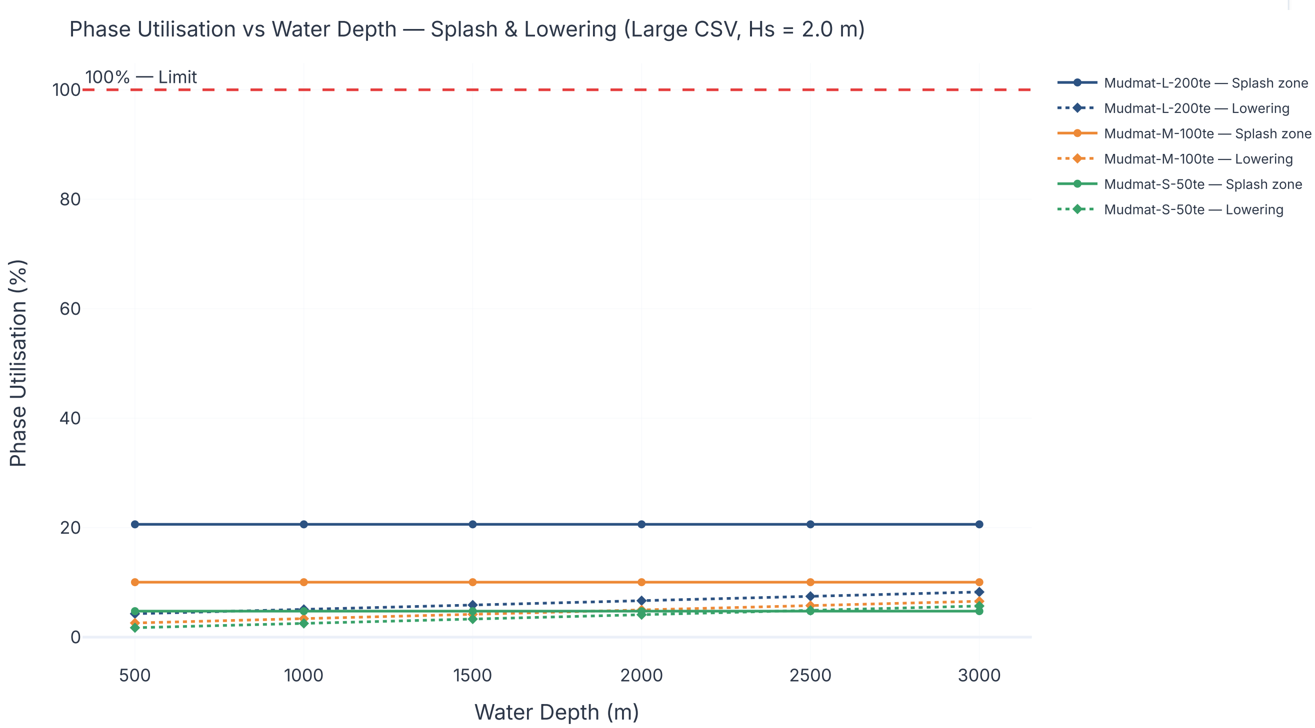


Chart 4: Operability Envelope — Max Hs vs Structure Weight

Maximum sea state where all depths pass. Defines weather window requirements.

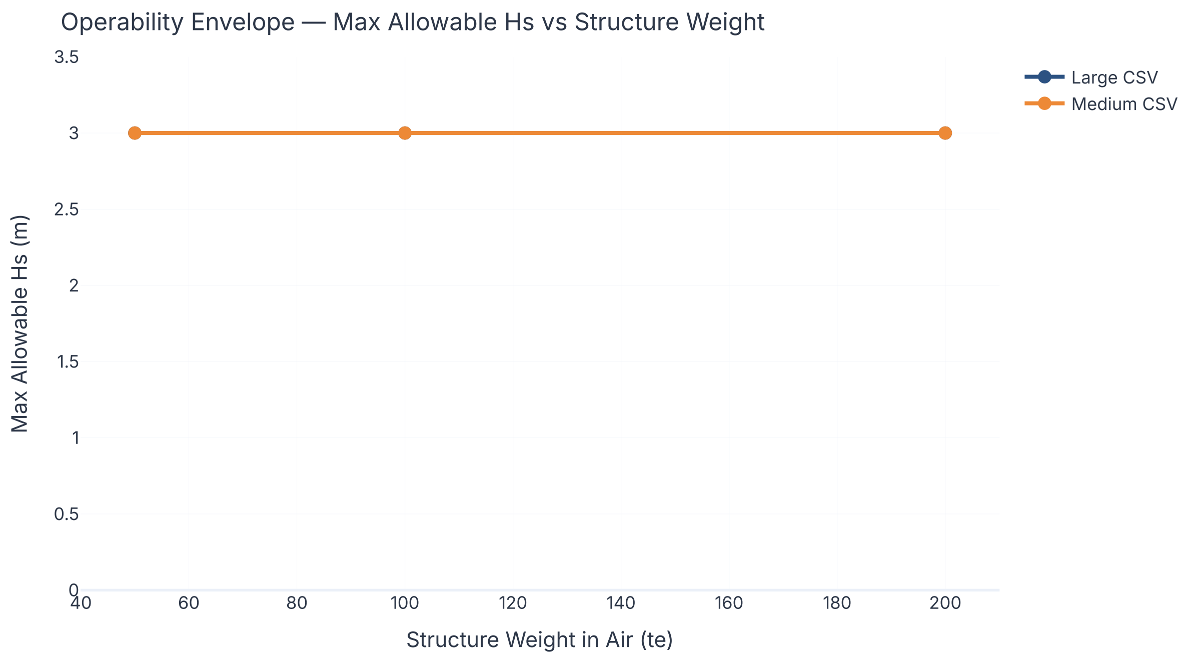
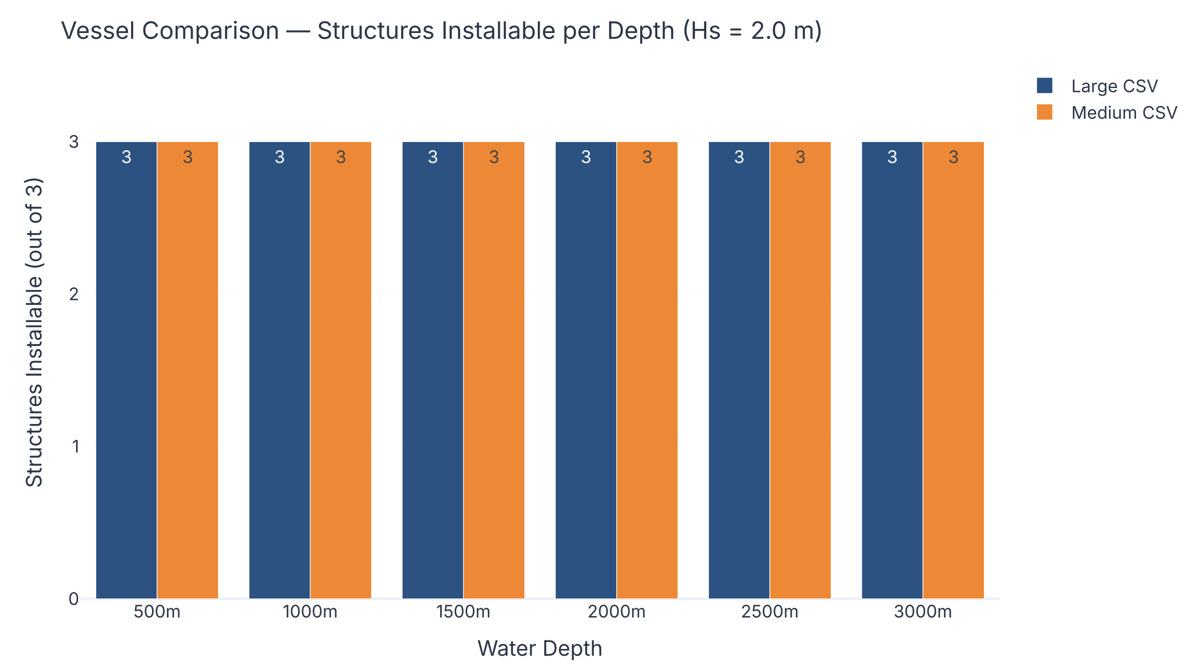


Chart 5: Vessel Head-to-Head — Structures Installable per Depth

Number of mudmat sizes (out of 3) each vessel can install at Hs = 2.0 m.



These results are preliminary engineering estimates generated by automated parametric analysis. All outputs require review by a qualified engineer before use in design or operational decisions.

References: DNV-RP-H103 (2011) — Modelling and Analysis of Marine Operations · DNV-ST-N001 (2021) — Marine Operations and Marine Warranty · DNV-RP-C205 (2010) — Environmental Conditions and Environmental Loads